

ALLPLAN 2023

New Features Allplan 2023

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Allplan 2023 – Your AEC platform to Design and Build together

The Multi-Material Solution for ultimate Buildability

Allplan 2023 is the connecting platform for interdisciplinary collaboration between architects, engineers, pre-cast plants and construction companies that delivers numerous innovations and improvements. As a multi-material solution, Allplan covers everything from masonry, cast-in-place concrete to steel and timber construction and – for the first time – precast concrete. The ability to coordinate different materials and construction methods in one common model enables architects to better consider the economical and sustainable use of building materials, earlier, and in accordance with environmental requirements. Engineers and construction companies can build directly on the architects' design and use it as the basis for structural analysis and detailing, MEP engineering, prefabrication and construction. Allplan supports workflows across the entire process from the initial idea to the completed project, enabling time, cost, and material savings.

Innovation and improvement highlights for architects include accelerated processing of terrain survey and point cloud data for refurbishment projects, faster modeling of complex openings and real-time BIM collaboration

between Allplan and Solibri. Engineers will particularly benefit from time saving new developments for automated detailing of reinforcement and new structural steel connection functionality. By integrating the previously separate precast solution Planbar, efficient design of structural elements including stairs, columns, slabs, and walls is now possible, together with automated plan creation. Construction companies will find time saving new features for excavation shoring and easier and faster site setup through powerful new functions, facilitating precise planning and therefore a smoother build phase.

Allplan 2023 provides a comprehensive platform for the entire construction industry, enabling efficient coordination and collaborative workflows between AEC professionals. The multi-material approach marks a key milestone in ALLPLAN's design-to-build strategy.

General New Features

A large number of new features at the various points of user management further increases the efficiency in daily work with Allplan.

More speed for download and installation

Allplan 2023 determines in advance which program components (functions, languages, country-specific settings) are required by the user. Only these are downloaded and installed. This noticeably speeds up the first program start. In addition, country-specific settings are now available for Canada and Mexico.

Acceleration of installation

The installation of Allplan has been considerably accelerated, by determining the program parts required for the installation before starting the download, and then by only downloading the installation package required for this.

Country-specific new features

Canada

The **English** and **French** languages as well as the **Fixtures, Textures** and **Vendors_FeatureHeadline** for the respective country are available for installation with the new **Canada** country code (requires a license).

Mexico

The **Spanish** language as well as the **Fixtures, Textures** and **Vendors_FeatureHeadline** for the respective country are available for installation with the new **Mexico** country code (requires a license).

Restoring drawing files directly from a backup

If you have a backup of project data either as a project backup (ZIP file) or as a copy of the project folder, you can restore individual drawing files for the project in question in a targeted manner without having to go via **ProjectPilot**.

To do this, select the **Building structure** tab in the **Open on a Project-Specific Basis** dialog box and open the shortcut menu for the relevant drawing file. Here you will find the new **Restore drawing file from backup tool** function.

Important! The function can only be selected if the drawing file is *disabled* fully.

You can find more information in the Allplan Help under "Restoring drawing files from a backup".

Relocation of the Allplan Exchange Online service

We are revising and modernizing the current Allplan Exchange Online service. The new Allplan Exchange Online service is due to go online after the Allplan Version 2023 has been released.

As a registered Allplan Exchange Online user, please ensure you read the information we have sent.

Performance – Confident handling of large volumes of data

Due to increasingly complex construction projects and the trend towards ever more detailed planning, data volumes are rising continuously and require correspondingly powerful software. Terrain surveys or point clouds with many millions of points represent a particular challenge. **Allplan 2023** processes such data directly on the graphics card. The data is not only displayed visually, but it is also possible to snap and measure points. In this way, large volumes of data can be reliably processed without time delays or memory limits.

Greater precision thanks to scalable user interface

Working with different monitor sizes and resolutions is a natural part of everyday planning. The right scaling is essential to achieve the highest possible accuracy and readability. As this is not always guaranteed via the operating system, **Allplan 2023** now automatically ensures that all elements such as menus, icons, palettes and graphics windows are scaled to the full screen resolution. This allows users to work precisely down to the smallest detail when modeling or visualizing. In addition, the icons have been made clearer. All in all, **Allplan 2023** presents itself with improved ergonomics and an attractive appearance.

Improvements in the Actionbar

Revised search in the Actionbar

The  **Find** function on the right side of the **Actionbar** has been revised and extended. The dialog box has been designed for ease of use and the search process is now faster. There are now also a number of functions which are not included in the **Actionbar** roles, but can be found in the  **Actionbar Configurator**. These functions are displayed in the **Find** data-entry box with the function icon and function name and can be started directly from here by clicking on the function icon.

Simpler access to SmartPart functions

The following functions used to generate door and window sill SmartPart can now be controlled as individual functions in the **Opening Elements** and **Interior Finish** task areas:

-  Sectional door SmartPart
-  Lifting door SmartPart
-  Up and over door SmartPart
-  Window Sill SmartPart

Plug-ins task in each role

The new **Plug-ins** task can be found in all of the roles. You can now use the **Actionbar Configurator** to save your required Add-On modules, such as **Lumion LiveSync**.

Label and design in engineering

In the role,  **Engineering** both of the **Label** and **Design** tasks are now available by default. You no longer need to switch to another role.

TT Reinforcement In the Engineering Structures task

The **TT Reinforcement** task area is now included in the **Engineering Structures** task by default.

Modeling – Model openings faster and more flexibly

Building models contain a large number of openings for windows, doors, facades or technical equipment. Special openings such as corner windows and wall or slab recesses, as well as their sub-objects such as lintels, rabbets or facings, can now be adapted even more quickly, conveniently and flexibly in terms of dimensions and representation using the properties palette.

Lintel, Header function – Surface elements

In the dialog box of the  **Lintel, Header** function (**Components** task area), you can now choose from all kinds of architectural elements (Hatching, Pattern, Fill, Bitmap Area, Style Area) and surfaces in the **Surface elements** area, just like you can for other architectural elements, such as  **Column**.

Attributes easier to adjust and localize

Attribution is of central importance for BIM-compliant design. With each Allplan version, workflows are therefore further optimized and made more user-friendly. Current innovations include user-defined property sets that can now be defined directly in Allplan. To improve clarity, imported attributes are displayed in groups. For simplified localization, formulas are now language independent.

Definable attribute-set-templates on office and project-specific basis

Attribute-set-templates can now be adjusted on an office and project-specific basis.

To do this, select the relevant project and open **Project Settings** (**New Project**, **Open Project** dialog box → Shortcut menu → **Properties**). Under **Attribute-set-template**, you will find all of the required functions to copy an existing attribute-set-template and then adjust it on an individual basis.

Alternatively, you can set the **Attribute-set-template** to **Project**. The attributes assigned to the project until then are retained for the project. You can also individually adjust the automatically generated project-specific attribute-set-template using the same method as copying an existing attribute-set-template: With  **Manage attribute set templates**, you open the shortcut menu by clicking on **Edit**. This opens the **Edit-attribute-set-template** dialog box in which you can make the corresponding adjustments.

You can find more information in the Allplan Help under the "**Edit-attribute-set-template**" dialog box.

Dimensioning views and sections in no time at all

During the course of a project, countless views and sections are generated from the building model. To make this process as time-saving and precise as possible, extensive optimizations have been made: For example, views and sections can now be created perpendicular to any surfaces or automatically with dimension lines. There are numerous new options for visibility and labels. In addition, many properties can be changed for multiple views and sections using the properties palette to save time.

Extended function scope

The functional scope of the existing **Sections** task area was expanded and renamed **Views and Sections**. You can now also use the  **Section Along Curve** function with the associated Dimension and Modification function.

The names of the parameters in these functions were adjusted to those of the  **Generate Section** and  **Generate View** functions.

Views perpendicular to reference surfaces

You can now define the viewing direction of views perpendicular to any reference surfaces. Once you have selected the  **Generate View** function in the input options, select the **Area** option and click a reference surface in order to do this. If the defined viewing direction corresponds to a standard view, this is configured in the **Views** area.

It is now easier and more intuitive to define a free view too. All you have to do when in the entry options is to select the **Free** option and click on a window to match the view displayed here.

New presentation parameters

The settings in the **Presentation** area were amended and expanded in the **Generate / Modify View** and **Generate / Modify Section** palettes.

- The name **Reference scale for calculation** was renamed to **Reference scale for 2D and 3D foils** in the  **Generate View** and  **Generate Section** functions to highlight what this setting does.

The value configured here is also used on the one hand for the presentation of smart symbols and fixtures which have foils with different scale ranges for a single view type, as well as for the reference scale for the drawing file if you generate the view using the building structure, or if you save your result on a different drawing file than the one that is currently active.

- With the new **Basic reinforcement precast elements** option, you can define whether the longitudinal and cross bars of the basic reinforcement are created as passive elements to reduce data volume and increase performance of precast elements. If this option is disabled, the reinforcement will indeed be drawn but it cannot be modified in this state.

Note: Manual presentation changes for an "active" reinforcement (such as placement display amended to  **Show middle bar only** or Dimension Line added) are lost when it becomes passive.

- With the new **Fixtures as wireframe model** option, you can define whether to show the fixtures as a wireframe model, so that you can see the fixtures better.

Note: Setting the **Presentation as a wireframe model** option under  **Options** now no longer has an impact on the presentation of the fixtures.

Extended text parameters

The text parameters have been expanded for Label for clipping path and Views and Sections.

You can now define a **Font-label**, specify the **Angle for italic text**, define the **Text color** and the **Layer**, define a **border around the text**, add a fill to the text and specify whether the size of the text on the layout will remain the same at all times, even if you change the scale of the layout element.

The extended text parameters are also available in the  **Section Along Curve** function for the section labeling and representation of the clipping path.

Favorites for formats

In the **Formats** palette, where you confirm the settings for the Hidden Line Image, you can now use the icons in the palette to confirm the formats for an existing view or section, save and load the settings as a favorite and reset the settings to the basic settings.

Position of heading and scale

In the **Labeling** palette, you will now see the new **Position** area when you generate the **heading** and/or the **scale** as the view/section labeling.

Here you specify the position and offset of the labelling in relation to the extent of the view/section. If you generate the **heading** as well as the **scale**, you can also define the **line spacing** for both of these headings.

The new **Position** area is also available in the  **Section Along Curve** function for the section labeling.

Automatic dimensioning

The new **Dimensioning** area is now available in the **Labeling** palette, both when creating as well as modifying views and sections.

Here you can specify whether the view/section is executed without dimensions (**off**), with  **dimension lines** or with  **elevation points**. If you have selected a dimensioning type, you can specify the parameters for the **dimension lines** or **elevation points** by clicking the **Set** button. You will already have seen the dialog box that opens from the dimensioning of reinforcement views.

Resetting the Section Identifier

In the  **Modify Clipping Path** function, you can now move a previously offset section identifier back to its default position by clicking

the  **Reset section identifier to default position** button within the palette.

Multiple modifications

With regard to the  **Remove/Add Elements** and  **Show Clipping Path** functions, you can now select several views and sections and modify them in a single step.

Reinforcement views only available to a restricted extent

Previously, there were two options for creating the general arrangement and reinforcement drawing. Of particular benefit for existing customers was the fact that they were able to work using the usual functions. However this was particularly confusing for new customers as there was no clear method. As stated previously, it is for this reason that the **Reinforcement Views** task area is no longer available for the general arrangement and reinforcement drawing. The functions in this task area can only still be used for the Element Plan and the Formwork elements of the role.

Although the way you work with the existing **Views and Sections** task area requires a little adjustment, virtually the same functionality remains in place. If you have only worked with the **Reinforcement Views** task area to date, you can start off by downloading step-by-step instructions contained in **Steps to Successful Views and Sections** from Allplan Connect, the service portal from Allplan.

Data conversion

If you load data or projects in Allplan 2023, elements created using the **Reinforcement Views** task area will be converted. As a result, the elements of the general arrangement views and reinforcement views are 2D design entities, texts and dimension lines. The format properties remain unchanged. Consequently, everything still looks the same in documents and layouts.

As the model data remain the same, although the existing linked layouts are no longer available, you can use the **Views and Sections** task area to create additional views and sections where required. The fully automated management of the number of pieces remains in place in any case.

Increased productivity through automated reinforcement

Reinforcement detailing is characterized by high work-load and deadline pressure. Automated reinforcement helps to reliably complete recurring tasks in less time. With **Allplan 2023**, it is now even possible to reinforce several columns and walls of one type at the same time. In addition, there are numerous improvements in detail. For example, columns can now have multiple stirrup types and continuous beams can also be reinforced in one work step.

With regard to the increasing requirements in BIM projects, the attribution of reinforcement has been extensively revised. The new release manager supports the cycle-accurate ordering of reinforcement in lean construction projects. In addition, there are numerous improvements in detail, for example, several polygonal placements with different mark numbers can be combined into one polygonal placement with one mark number.

Extensions in automatic reinforcement

The functionality to create automatic reinforcement for beams, walls and columns as well as the automatic punching shear reinforcement was expanded and improved further.

- Beam reinforcement for multi-span beams possible.
- Mesh reinforcement and spacers for walls.
- Multi-section stirrup reinforcement for columns.
- Create multiple walls or columns in one step.
- General improvement of assigning layers and creating sections.

Additional criterion when rearranging

The Rearrange Marks function was expanded to include the **Prefix** criterion. When you select this option, the **Rebar Prefix** attribute will be taken into account when rearranging the marks. Identical reinforcing bar marks with different attribute use are not therefore combined into a mark.

Coupler parameters modifiable

The  **Modify·Coupler, Thread, Connecting·Bar** function now provides you with two new functions that you can use to amend the parameters for couplers or threads via the pull-out bars or placement. Once you have clicked on the coupler or thread, the parameters set as visible in the Article editor are then displayed in a dialog box.

You can modify parameters for which a list box opens by clicking twice in the **Value** column, such as for the new end processing abbreviation. The remaining parameters are purely for information purposes.

Labeling PythonParts

As is the case for other objects in Allplan, such as SmartParts, you can now open the  **Label** function in the shortcut menu for PythonPart objects.

Other New Features

- The  **Extrude Bars Along Path** and  **Sweep Bars Along Path** functions are now also available for PythonPart development.
- You can now use  **Join bars** to edit polygonal instances too.
- The reinforcing steel processing guideline is now taken into account for Switzerland.
- The standard settings have been expanded for the USA. Abbreviations can be assigned for the end processing of couplers and threads, and the Release Manager is also available to order reinforcement.

Efficiently Convey Constructability with Design Intent

Users now have the ability to customize the Start and End Angles of Structural Framing Elements, which is necessary to accurately model steel structural framing situations. The Connection Toolbox now includes the ability to specify Holes and End Conditions. These features help Engineers model most Steel Connections at LOD 300 and above, improving accuracy and efficiency of communication while collaborating with detailers, fabricators, building engineers, and other BIM stakeholders.

Furthermore, the Connection Toolbox is written in Python and can be extended by users and developers. Column Splice Plate and Base Plate examples have been shipped with **Allplan 2023** to help get started with extension development.

Connection of structural framing beams and braces

With the  **Join Linear Component with Line** function, you can now also connect structural framing beams and structural framing bracings to the lines or edges of other Structural Framing Objects. You will also find this function in the shortcut menu for both of these Structural Framing Objects.

After you have chosen the required Structural Framing Object, you then go to the input options to define whether it is just the anchor point to be connected to the axis of the Structural Framing Object, or the entire Structural Framing Object together with the anchor point and axis. If you select the **Adjust final outline** option in the second instance, the inclination of the cross-sectional area will also be adjusted to the line or edge.

Inclined cross-sections for structural framing objects

When you are in the palette for defining the Structural Framing Object parameters, you now have the option on the **Geometry** tab to define an angle in the local x- and y-direction between the cross-sectional area and the perpendicular to the object axis at the start and end of the Structural Framing Object. The cross-sectional area is always rotated here by the anchor point.

If inclined cross-sectional areas arise when a beam or brace is connected, or if inclined cross-sectional areas arise for inclined beams or braces owing to the selected beam end, the corresponding angles will be entered automatically.

Road Design – Productivity boost for terrain and road construction plans

To boost productivity on infrastructure projects, the new version improves and automates plan rendering and introduces additional components and layers. The reduction of manual workflows leads to significant time savings. Smooth data exchange is supported by an IFC road-compatible building structure and attributes. Road construction works can be evaluated on the basis of national regulations with the advanced testing routines for Germany, Austria and Switzerland.

Extensions for the road construction

The description of the functions in the **Terrain** and **Street** tasks of the role  **Road construction** is now integrated into the Allplan Help. The following extensions and improvements have also been made.

- Localization – country-specific standard testing
- Import and visualization of cadastral data
- Improved terrain display (point clouds)
- IFC 4.3 Standard for road objects
- EN: Axis import in OKSTRA format
- Horizontal geometry – transition arcs according to Bloss
- 3D dimensioning of the road body
- Reports – Output of peg points (x, y, z)
- Quick access to general settings

Civil engineering – Precise planning of excavation shoring

Precise planning of excavation shoring is particularly important for inner-city construction projects and infrastructure projects. **Allplan 2023** supports these tasks with new functions for planning bored pile and soldier pile walls, as well as ground anchors.

Advanced task areas for construction

In the role,  **Construction** the **Building site** task has been expanded to include the **Crane planning** task area. You will now find the **Tower crane** and **Capacity checker** functions here.

The **Site facilities** task area has also been expanded to include the new **Placement in grid** and **Placement along path** functions. You can use these to create containers or site fences, for example.

In the **Terrain** task, you will now find the new **Civil engineering** task area, which includes a number of functions to create excavation shoring.

Interfaces – Best possible OPEN BIM workflows for reliable collaboration

To provide the best possible support for data exchange with OPEN BIM and proprietary formats, the Allplan interfaces have been comprehensively updated. For IFC, DWG and DGN formats, Allplan now uses the current libraries of the Open Design Alliance, an association of 1250 companies in the construction industry. The reliable cooperation in openBIM projects has certified buildingSMART for the export of the IFC4 Reference View.

For interaction with structural analysis software, the SAF format has been defined, which is now supported by 16 software companies. Structural models from Allplan Bridge can be transferred to Midas. For road projects, export in IFC 4.3 format is now available. In addition, cadastral data with property boundaries can be imported in SHP format. The new IFC4precast interface enables model-based data exchange that is independent of the production facilities used.

In addition, special developments were realized for various countries, such as support for XPlanung for Germany (availability planned for 4th quarter 2022), attribute templates in accordance with the Czech standard, and improvements in reinforcement export in aSa and Soule formats for the USA.

Improved export with resources

When exporting drawing files and layouts with resources, it has previously been the case that only all loaded drawing files (current – edit – reference) or the current layout with its drawing files were saved as an NDW file. If the loaded drawing files contain associatively referenced files that are not loaded, you now have the option to choose whether these files are also copied. If you have loaded Views and Sections without the associated model data, these data shall also be copied.

NID export of multiple layouts

With the  **Export NID File** function, you can now export multiple layouts in one step.

If you want to export other layouts in addition to the current layout or take other layouts into account for the export as well, click the **Layout number** button and select the required layouts via **Layout structure**. The suggested file name is formed using the layout name and the smallest layout number.

Extension for reinforcement export

With regard to the  **Export aSa** and  **Export Soule** functions, all bar reinforcement elements of the loaded drawing files are now taken into account, including drawing files in reference and edit state.

For the reinforcement export using the  **Export aSa** function, you can now define whether TEK or RDX files are to be created from the existing documents.

BIM in real time between Allplan and Solibri

Instead of importing and exporting BCF files, the connection of the Allplan/Bimplus* Issue Manager with the Solibri BCF Live Connector now allows issues identified during model checking in Solibri to be transferred directly to the issue workflow in Allplan. Thus, issues can be evaluated and processed, edited, and corrected directly in Allplan.

*Bimplus Pro project required. Bimplus Pro license included with Serviceplus or Subscription.

Documentation – Efficient work with text leaders

Text leaders are particularly helpful when a lot of information needs to be displayed in the same place. With **Allplan 2023**, the text leaders have been extensively revised so that, for example, moving or changing scales is now much smoother. It is even possible to use text leaders to label several objects simultaneously.

Text leaders immediately after text creation

By selecting the  **Text Leader** option in the advanced toolbar of the **Text** dialog box, you can now append a text leader to a fileset element once the text has been definitively created. The  **Text Leader** function is automatically started for this purpose.

If you wish to edit text, you cannot select the  **Text Leader** option.

Multitext leader

You can now use the  **Text Leader** option to create multiple text leaders for text.

After you have generated the first text leader in the usual way, you must then select the  **Sub-text leader** setting. Where required, change the setting so that the text leader is generated as an  **individual line** or as a  **polyline**, then change or define the start and/or end symbol. Finally, click the text leader in the position where the additional leader is supposed to start and enter the text leader.

Correct text leaders for each modification

If you move text, copy it across multiple drawing files, modify parameters such as text height and width or change the reference scale, the presentation of the text leaders will always be updated correctly.

Text wrapping for text modification

If you click on text without one of the functions being active, the text will now be provided with a border frame.

You can select the corners of the border frame and move them into specific halftoning. This then generates text wrapping for the text, i.e. the words in a line of text are wrapped in such a way so that the text fits solely within the border frame. If you wish to remove this text wrapping generated this way in full, click on the text and turn off the **Text wrapping** option in the **Properties** palette.

Visualization – Impress with perfect renderings

Convincing clients of creative ideas is often best achieved with atmospheric visualizations of the project that are as close to reality as possible, instead of 2D plans that require explanation. With **Allplan 2023**, the extremely powerful visualization options have been extensively enhanced. For example, visualization effects such as bloom and lens flare, depth of field and volumetric fog are now also available throughout the animation as well as with Realtime Render and CineRender. Newly added are further professional material settings. Interaction with CineRender during changes has been significantly accelerated. Additionally, the integration of the Lumion Live-Sync connection improves the visualization workflow by tracking Allplan scenes and their changes in Lumion in real time.

Model calculation in Fly mode

In addition to the known  **Design mode** and  **Navigation Mode** navigation modes, you can now also calculate your model in  **Fly-mode**. You can control the presentation in Fly-mode by pressing and holding down the keys on the keypad known from other applications (W – A – S – D):

- W – Fly to model
- A – Fly to the left
- S – Fly away from model
- D – Fly to the right

You can also use the middle and right mouse button as well as the ARROW KEYS on the keypad to control this.

Select  **Fly-mode** via  /  /  **Navigation-mode** in the window toolbar of a graphics window for each individual graphics window.

Reflection of rough and structured surfaces

When defining surface properties using the  **Set Surface** (**Surfaces, Light** task area) function, for example, you can now save a separate texture for **roughness** – in a manner analogous to **glossy reflection**.

The settings for roughness and glossy reflection will mutually impact each other. If you have selected intensity > 0 for the glossy reflection and coat of lacquer > 0, the coat will generate a glossy reflection without taking the settings for roughness into account. If glossy reflection = 0, a coat of lacquer > 0 will take the roughness intensity and roughness texture into account.

New features in Visual Scripting

The following new features have been incorporated into Allplan Visual Scripting:

- The toolbar has been expanded to include the  **Stop** function (including in the **Script** menu) and the  **Create objects in Allplan** function (including in the **File** menu)
- The  **Create PyP script only...** function has been added to the **File** menu
- You can now activate the **Watch list** palette in the **View** menu
- You can use CTRL+C and CTRL+V to copy and paste selected nodes across scripts.
- A project file ending *.avsprj is now created when saving a VisualScripting script

New functions in Visual Scripting

The  **Stop** (SHIFT +F5) function can be found both in the toolbar as well as in the **Script** menu. It ensures the execution of the script triggered by  **Start** and, thus, synchronization between Allplan and Visual Scripting is interrupted. If the script has already been executed and the Properties palette and the object are displayed in Allplan, switching the  **Stop** function on will ensure that this preview in Allplan is deleted again. You are therefore returned to the initial state without having to close Visual Scripting.

The  **Create objects...** function can be found in the toolbar and in the **File** menu. If you have created a script in Allplan Visual Scripting and triggered the execution of the script by selecting  **Start**, the object und Properties palette will be displayed as a preview in Allplan. To ensure both are not only displayed but also created in Allplan, you must use the  **Create objects...** function. The object is then saved in a *.pyp file. By doing this, the object and Properties palette will continue to be available for editing in Allplan after the Visual Scripting application window is closed.

The  **Create PyP script only....** function found in the **File** menu only allows you to create a script and not an object in Allplan. Your script is saved in a *.pyp file.

New functionalities in Visual Scripting

You can activate the **Watch list** palette in the **View** menu. However, it is also opened if you show the tooltip of a node parameter at the output ports in a node, click the  **Watch** button and add this parameter in the **Watch list** palette. The **Watch list** palette is docked at the right side of the application window. You can now check the parameters you have added for the duration of your Visual Scripting session. If you close **Allplan Visual Scripting**, the **Watch list** palette is automatically cleared.

If you select a parameter in the **Watch list** palette, the associated preview element will be highlighted in color in Allplan. This allows you to review the geometry of your element.

You can use CTRL+C and CTRL+V to copy and paste selected nodes and their connecting lines across scripts. You can reuse existing node combinations by copying scripts.

A project file ending *.avsprj is now created when saving a VisualScripting script.

If you have created VisualScripting scripts in previous versions of Allplan 2023 and then saved them, they will have been saved as a *.pyp file. In the same way as before, these files can only be read or edited. When saving these data, they are now saved as a *.avsprj project file.

When in the workspace of the Visual Scripting application window, you can now arrange for your script view to be enlarged or reduced by a certain percentage. To do this, you can choose from 7 preset values or use your mouse scroll wheel to set the presentation size.

Precast planning – automated precast planning and new precast element tools

The **Allplan 2023** version is characterized by the integration of numerous functions for precast planning. Specifically, the complete range of functions of Planbar, one of the most innovative planning solutions in the Precast area, has been included in Allplan. As a result, engineering offices and precast plants can now design precast concrete elements of any complexity directly in Allplan. Allplan automatically creates element plans for production and the identifier can be used to check whether precast elements with their specific fixtures and reinforcements are identical to other objects.

Merging Planbar and Allplan not only simplifies project coordination, but also enables new, even more efficient workflows. For example, precast elements can be created in a time-saving manner using functions in Allplan as well as PythonParts.

The joint version offers numerous advantages to existing Planbar users: They gain access to additional Allplan tools such as Bimplus or Allplan Share, as well as to additional functions for cost planning, civil structures and much mo-

re. The administration effort is also significantly lower, as installation and licensing are only required once.

Precast plants get sophisticated planning workflows for highly automated wall and slab production as well as numerous interfaces to production machines with the Allplan Precast product in addition to the range of functions in Allplan.

In addition to the integration of precast functions in **Allplan 2023**, there are other innovations in the Precast area. One highlight, for example, is the Smart Converter, which interprets and analyzes stair drawings (2D or 3D). It then corrects angles if necessary and transforms the drawings into producible, fully parametric stair models. Designing precast staircases has never been easier or more efficient.

In addition, the layout catalog has been extended by numerous setting options. As a result, the layouts for element plans can be designed even more individually.

Fitting parts can now be generated with Python scripts. This opens up completely new perspectives with regard to fixture functionality. Definable rules increase the degree of automation. For example, Allplan can automatically calculate the correct positions and diameters for necessary holes based on the dimensions of a steel plate. Parametric resizing is also easy to carry out.

New features across modules

Functions of the Reinforcement Views task area

As announced for version PLANBAR 2022-0-1, the functions of the **Reinforcement Views** task area for the placing drawing do not apply.

The functions are now only available in the shortcut menu of both functions for the  **Element Plan** and the  **Panel Processing** and in the plan view for elements of the **Formwork** task. In all other cases, the selection of functions is cancelled with a message. Use the functions of the **Views and Sections** task area instead for the placing drawing except for the **Formwork** task.

For the elements of the **Formwork** task, no reinforcement views or sections across drawing files can be created. For this reason, the  **Display Drawing File References** and  **Manage Reference Drawing Files** functions have been removed. If you try to create a reinforcement view or section on a drawing file other than the selected elements, the process is cancelled with the message **Reinforcement views across drawing files are no longer available**.

Note: The **Prestressing** task area has been removed entirely from the **Reinforcement** task of the  **Precast elements** role. As a result, the **Reinforcement** task now fully corresponds to the  **Engineering** role. Now select the  **Tendon** function in the **Engineering Structures** task area of the **Engineering Structures** task.

Separate task area and tab for fixtures with precast elements

In the Actionbar, the higher  **Fixtures** function and the corresponding functions in the Context Toolbar have been replaced by the separate **Fixtures** task area, which is also used in the  **Engineering** role in the **User-Defined Objects** task for the precast slab, precast wall and precast elements. In the **Fixtures** task area, you can then call up the functions directly.

Please note that the  **Fixtures** function has now moved to the first position of the **Create** function group and the  **Define Fixture** function is at the end.

Note: In the palette configuration, a separate  **Fixtures** tab has been added as a replacement for the  **Fixtures** function after the tabs for the precast slab, precast wall and precast elements.

Cutting the reinforcement and Interaction with precast element for Model Precast Elements

If a linear fixture is selected for  **Model Precast Elements** as **Fixture for Invoicing**, for which options for cutting the reinforcement are active under  **Catalogs**, **General information**, **Fixture catalog** on the **Reinforcement** tab, these options now fundamentally remain overlooked. The fixture in question is only used as an invoicing component.

Furthermore, there is fundamentally no further reinforcement cutting in  **Model Precast Elements** on fixtures for which the **Positive (merge) / lengthening** option is set to **Interaction with precast element**.

Presentation of bending shape in plan view or on element plan

In the floor plan and on the element plan, the bending shape symbols were previously drawn exactly on the support edge when entering **bar overlap = 0**. However, when creating the bars, the bars were shortened by the configuration setting **Fixed value (bar overlap = 0)**. With the change in Allplan, the bending shape symbols are now also offset by this value.

Note: This applies only to new placements and to existing placements where the span direction or the supports involved are changed. For existing placements with no change, you will still have the old representation.

For wall element designs created using the  **Design (iWall)** function, the bending shape symbols of the **Basic reinforcement** (for each reinforcement unit) at element edges are now shifted by the **Lateral concrete cover** when you enter a **Bar overlap = 0**. For recess edges, the **Lateral concrete cover** (per reinforcement unit) applies, which is set on the **Reinforcement of openings** tab.

PythonPart crane from the library

The load radii – in the **Library** palette of Allplan in the **Standard-resolution** with the **PythonParts** under **Construction** – with the crane delivered are now also just as recognised as the load radii of the cranes created with  **Crane Location** in the **Precast Slab** task and **Precast Wall** of the **Precast-elements** roll.

Note: Define one or more crane locations for the crane as Python-Part before the design, so that when running the design function the crane parameters are taken into account when calculating the individual elements.

New attribute thickness ID

For use in alternatives text images, title blocks, legends, etc., you will now find the new **Gauge Mark** attribute (@36098@, String) in the default attributes under the Precast Elements, **element-specific** attribute group.

As a result, the value defined on entry or modification for entry of the **Gauge mark** parameter in the **Labeling** group under  **Design (i-Wall)** is output with the settings for the **Design Mode** on the **Attributes** tab.

Catalogs and Configurations

Production file name for production or ERP data

The **Production file – Name convention** defined up to version PLANBAR 2021 in the now omitted  **Configuration, Site facilities, General, Production Data Transfer, General** on the **Program Flow** tab can now be defined again. A complete elimination of this parameter was not possible, since it is used by many customers simultaneously with the export file name, as an alternative way to create names (e.g. to control different systems that require different name patterns).

Therefore, a second export name has now been provided under  **Configuration, General, Program Flow, Export name** on the **Export file name** tab. This second name is called **Production file name for production or ERP data** for pragmatic reasons, so that the former setting from the configuration is easier to assign. The entries for this are made in the same way as the other settings on this tab.

When the configurations are opened for the first time, the existing configuration under **Production file – Naming convention** from  **Configuration, Site facilities, General, Production Data Transfer, General** on the **Program Flow** tab is converted to the new setting.

There are no changes for the  **Production Data, NC Generator** and  **List Generator** functions. You can continue to freely decide here whether the **Export file name** or the **Production file name** should be used.

Note: Please check the correctness of the attribute numbers used after the update. There are two possible attributes for the floor in Allplan that were used earlier. **Slab above** (@829@, string) and **floor** (@1092@, string). If **floor** was not used, **slab over** was used. This now no longer works, because at the time of the conversion of the setting it is not clear which attribute is in use. Therefore, the attribute **floor** (@1092@) is entered during the conversion. If the attribute **slab over** is used, please replace the attribute number with @829@!

Catalog reference for fixture group, group leading

Since PLANBAR 2020 there has been a **Group leading** option for defining non-dynamic fixture groups. In a non-dynamic fixture group with the **Group leading** option, all fixtures in Allplan are active together. They are no longer deleted or moved individually. All related fixtures are deleted, moved, etc. All fixtures belonging to the group are linked to the same element and, if applicable, to the same precast element layer, even if individual parts lie geometrically in another precast element or another layer.

The new version Allplan 2023 now also offers the option to individually assign a catalog reference to group definitions. As a result, it is possible to evaluate and treat the group itself as a fixture. The individual parts can then, depending on the setting, also be evaluated or omitted. Since the fixtures of a leading group are always linked to a single layer, even if they lie geometrically in another layer, it is not possible to calculate the respective page (**S1, S2, U1, ...**) and this is no longer written for those fixtures.

- If a non-dynamic fixture group with the **Group leading** option is installed without reference to an entry in the **Fixture catalog**, all fixtures of the group are linked to the leaf in which the reference point lies. Geometrically, however, they lie in different leaves. The surface labels in the table on the element plan are created in the leaf to which the fixtures are linked. Since the group definition has no fixture reference here, the individual fixtures continue to be handled as before.
- If the non-dynamic fixture group with the **Group leading** option assigns a reference to an entry in the **Fixture catalog**, this affects the labeling in the table on the element plan. Now the name of the group is written first before the fixtures contained in the group.

If individual parts of the group are not listed in the table, this must be adjusted in the **Fixture catalog**. The **Surface labels in the table** option on the **Element Plan 1** tab must be disabled on the corresponding parts.

If a dynamic fixture group appears several times with the **Group leading** option, the number of groups is summed accordingly and written to the table on the element plan. The number of fixtures

contained in the group is, however, only written by one group. Groups are combined when all components of the group, which are displayed in the table, are identical. If only the group is displayed (without the individual components), only the groups are compared. If only one component of the group is displayed, the group and the components are compared.

A dynamic fixture group with the **Group leading** option is listed in the  **List Generator** in addition to the parts of the group (according to the settings in the **Fixture catalog**). In other words, the group and the individual components are recorded in the lists. If multiple elements exist, both the groups and the individual parts are summed, unlike in the element plan. If individual parts have no catalog reference, these are not listed. This is the simplest way of suppressing individual parts in the lists. However, the results may be unwanted in other modules.

Note: If individual components or the entire group are not listed – in the **Fixture catalog** on the **Logging** tab in the **Logging** group for **Involving Customer** and/or **Involving Factory** – a **Logging formula can be used** that returns **0** as the result when calculating the value for the individual fixture. As a result, the parts without effects on other modules can be filtered accordingly.

Attention: Since structuring of the outputs is not planned in any of the returned standard lists, the output occurs on one plane and possibly in another sequence. This is justified by the fact that the fixtures are sorted by list and therefore no longer have any correlation. Moreover, fixtures are also recorded separately by type on the lists (symbol, line or surface fixture or formula fixture).

If a catalog reference is saved with the **Group leading** option for a non-dynamic fixture group, only the group is still transferred as the individual fixture when creating the production data with  **Production Data, NC Generator** in the file for the NC generator and the PXML file. However, this fixture can display multiple geometries depending on the settings in the **Fixture catalog** on the **Production** tab.

- Only the geometric attributes are evaluated on selected parts of the group, such as how the individual part is to be presented (**Presentation such as CAD Foil**, replacement symbol, etc.).
- All other settings such as **Consider projection** or **Additional Information** depend on the catalog reference of the group

- If an individual part has no catalog reference or if **Create fixture data = no** has been set, the geometry is "passed" to the group.
- When choosing a substitute symbol and selecting **Calculate max. expansion**, only the geometry of the part is considered for each part of the group. If a substitute symbol is set for the entire group, however, and **Calculate max. expansion** is set, the min-max box of all parts is taken into account.
- The **No – Presentation such as CAD Foil** option is only possible on individual parts. If this option is set for the group, data is presented by 3D solids of the transferred geometries of the parts (if applicable).
- A reference point of all parts with the lowest XY coordinates, as a reference point for the group.

Non-dynamic fixture groups with the **Group leading** option, which are exported in link with precast elements in the IFC, are exported automatically as **IfcElementAssembly**. The **IfcElementAssembly** is automatically given the list text set in the catalog as the name. The attributes written on the group fixture are exported to this assembly. The fixtures installed in the group are output in the IFC also as parts of the assembly.

Note: For the special scenario of an Iso basket as a non-dynamic fixture group with the **Group leading** option, nothing has been changed for reasons of compatibility. This means that here no catalog reference to the group definition should be saved, as special treatment occurs in the annotations for the Iso basket, which would no longer pass.

Should you have any further questions on this, please contact your personal sales engineer or our technical support team in Puch.

Insulation Material Catalog, deduction value for cutting tool

Two values have always been defined for the **Half-panel-length** and the **Half panel width** in the **Insulation Material Catalog** under  **Catalogs, General information** on the **Placement** tab, however, these are only ever taken into account within the program.

This has now been changed. An individual **Subtraction value for cutting tool for half slabs** is now always defined to take account of the width of the cutting tool

Layout catalog, new functions for formula editor

When using the formula editor to define an iTrigger or labeling for the layout catalog, the **OPENINGMACROCOUNT** and **GROUP** functions have been extended specifically.

- Using the new **OPENINGMACROCOUNT** function you can, for example, count all openings with an opening macro for slabs or wall elements (iWall), which geometrically influence the selected precast element. Openings that lie outside or that only touch the precast element are not counted. Other criteria can also be used in the same way as the **FIXTURECOUNT** function.

For example, the formula **OPENING-MACROCOUNT(@507@;"Holzab")** counts all openings containing a macro with the **Name** (Attribute **@507@**) = **HolzAb**.

- Use the **GROUP** function to group elements with attributes, such as reinforcement, precast elements, etc., (in other words, combined into groups). 'As a result, almost identical dimensioning can be achieved for the reinforcement bars, for example, as for precast elements in combination with the **Dim. string index**.

To do this, use the desired elements automatically or also manually (e.g. reinforcement bars) with an attribute. Next assign a value for this attribute with  **Assign attributes**.

Depending on how the elements (e.g. reinforcement bars) are to be grouped, a formula must then be adapted with the **GROUP** function so that all components with identical index are grouped together on the same dimension line. You can also group together

multiple indices on a single dimension line.

The **GROUP** function considers 3 different scenarios:

- Attribute on considered element (e.g. bar) available & in group; bar dimensioned & grouped
- Attribute on considered element (e.g. bar) available & not in group; bar not dimensioned
- Attribute on considered element (e.g. bar) not available; original dimensioning for bar

For example, one grouping using the attribute **Dim. string index** (@1013@, Integer) could appear as follows; **GROUP**("1-4:A; 5-8:B";@1013@). All rods with dimension string index = 1-4 are listed on a dimension line, which receives text A as dimension line name. The rods with dimension string index = 5-8 are listed on a second dimension line, which receives text B as dimension line name.

Note: Only one individual attribute can ever be evaluated with the **GROUP** function.

Should you have any further questions on this, please contact your personal sales engineer or our technical support team in Puch.

Layout catalog, dimensioning of bricks

For a brick arrangement with a **placing angle** of 0°, 90°, 180°, 270° or 360°, as many dimension lines are always generated as there are different rows for the dimensions of rows of bricks in the Element Plan for the walls generated using the  **Design (iWall)** function where **Element type** = **Brick wall**. In the case of non-staggered laying, one single dimension line is drawn; in the case of staggered laying, two dimension lines are drawn.

Note: In order to dimension the bricks in a view/section, go to  **Catalogs, General, Layout catalog** on the **Dimensioning** tab, then **Special components** and choose the option **Brick and tile areas/bricks/tiles**. Or go to the **Shell/Presentation** tab and choose the option **Bricks/tiles** under Presentation.

If additional brick areas are added in  **Modify Wall in View** with  **areas for surfaces, concrete, insulation, brick and tiles**, they will be dimensioned on their own dimension lines.

Layout catalog, dimensioning of hollow blocks in walls

The hollow blocks in the walls generated using the  **Design (iWall))** tool where **Element type** = **Concrete wall** and **Sandwich wall** are now dimensioned on their own dimension lines.

To do this, select the new **Special dimensions for hollow blocks** option under  **Catalogs, General, Layout catalog** for views with **viewing direction** = 1/6 or 2/4 on the **Dimensioning** tab in the **Geometry** group. You can decide whether the **min-max points** or the **center of gravity** is to be dimensioned on one dimension line per hollow block (= **Dimension separately**) or on combined dimension lines (= **Dimension together**).

Note: Under  **Catalogs, General, Layout catalog** in **General, Dimension lines, Dimension groups**, you can customize the default HO designation for the dimension line(s) of the hollow blocks.

Layout catalog, change for panel edge dimensioning settings

The options for **panel edges** and **connections** in a view or section under  **Catalogs, General, Layout catalog** on the **Dimensioning** tab in the **Geometry** group have been revised.

The previous selection dialog with the options **No**, **Only panel edge** and **Panel edge and connections** for the parameter **Panel edge** has been changed into two settings for **Panel edge** and **Dimension connections** with check boxes (Yes/No).

If you disable **panel edge**, the other parameters will not be displayed. Only when you activate **Panel edge** can you specify whether the options **Dimension together**, **Dimension separately** or **Dimension hidden edges separately** are to be used for **Dimensioning panel edge**.

The **Dimension separately** option has been completely revised, as this setting previously gave the same result as the **Dimension together** setting.

Following this revision, more flexible settings for the precast element layers are possible with regard to the precast element dimensioning. Connecting points can only be ignored or dimensioned for individual layers. It is possible to have separate dimension lines for the individual precast element layers, which can therefore be easier to identify as a result. This helps reduce the complexity of the points to be dimensioned on the panel dimension line.

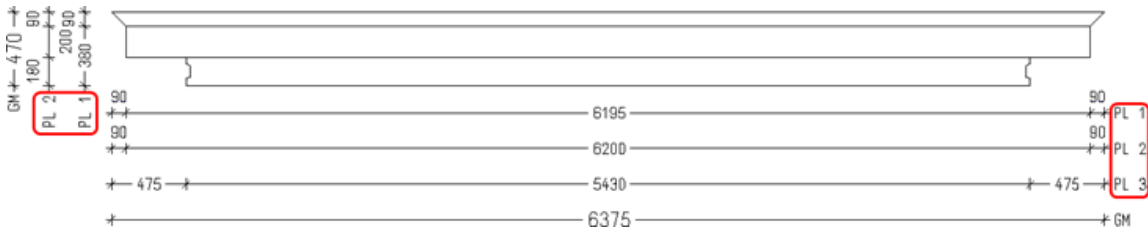
For compatibility reasons, a set option **Dimension separately** is now automatically set to **Dimension together**. The entire precast element is now dimensioned on the panel edge dimension line. The effects of the **Dimension hidden edges separately** option remain unchanged.

If you activate the revised option **dimension separately**, you can make special settings for dimensioning the layers. A selection for the component **Layers** appears first. ◀ and ▶ take you from entry to entry. With , open the sub-dialog where you can make the settings for all layers in a detailed dialog.

Here you can make the following configuration:

- **Dimensioning**; specifies whether the layer should be dimensioned or not.
- **Dimension separately**; specifies whether or not to create your own, separate dimension line for the layer. If you select this option, the number of the layer is written to the dimension line so that you can identify the dimension line.
- **Dimension connections** (only for walls); specify whether points of the connection that are in the layer should be dimensioned or not

The setting **Dimension connections** is now only visible for the layout for one wall. This is not required for slabs, structural precast elements and precast elements, as no connections are possible here.



The following options are visible in the example below:

- All layers of the triple-layer precast element are shown, although only some of the individual layers are dimensioned.

See, for example, the lateral dimension lines for layers 1 (PL 1) and 2 (PL 2) to the left next to the panel and the dimension lines for layers 1 (PL 1), 2 (PL 2) and 3 (PL 3) below the panel.

- The connecting points in just their individual leaves are ignored or only dimensioned in individual layers.

For example, see the connection in layer 3, which contains a nut. This is not dimensioned in the lateral dimension line. The dimension line for layer 3 is omitted.

Note: The option **dimension separately** option is no longer available for precast elements of the **structural precast element** type, because there are no layers here.

Layout catalog, change for opening and recess dimensioning settings

The options for **Openings and Recesses** in a view or section under  **Catalogs, General, Layout catalog** on the **Dimensioning** tab in the **Geometry** group have been revised.

When selecting the **Dimension separately** option for **opening and recess dimensioning**, the **Dimension mode** additional parameter is now displayed.

Here you can define whether the openings or recesses should be dimensioned either by the **Min–Max Points** or via **All points**. This will generate different results for trapezoidal openings or recesses in particular. The default setting for this new option is **Min–Max Points**.

Layout catalog, mark number and dimensioning of fixtures

The setting **Label fixtures** or **Dimension fixtures**  was added under **Catalogs, General, Layout catalog** in the **Properties** for the views and sections. They can be found in the **Text** tab under **Fixtures mark number** or in the **Dimensioning** tab under **Fixtures**.

You can use **All**, **Only these** and **Not these** to define which fixtures are to be labeled with a mark number or dimensioned.

If **All** is selected, the dimension line index is not used as the basis for labeling nor dimensioning. If **Only these** is selected, only the fixtures with the configured dimension line indices will be labeled with the mark number or dimensioned. If **Not these** is selected, all fixtures, except those with the configured dimension line indices selected, will be labeled with the mark number or dimensioned.

Layout catalog, labeling of symbol fixtures on the dimension line

The labeling of symbol fixtures on the dimension line is now active for the absolute dimensioning.

To do this, select the **absolute** option under  **Catalogs, General, Layout catalog** on the **Dimension lines** tab under **Horizontal** and/or **Vertical dimension lines** for the **Dimension** parameter.

Then select the **As per labeling** option on the **Dimensioning** tab in the **Fixtures** group under **Special dimensions for fixtures**. For **Dimensi-**

on lines **additional text**, select one of the **horizontal**, **vertical** or **horiz+vertl.** options and define the **template** for labeling. You can use the Allplan **Formula editor** here to define additional text as the labeling for fixtures on the dimension line.

Note: Please note the **Dimensioning Point(s)** setting must be selected for the corresponding fixtures under  **Catalogs, General** in the **Fixture catalog** on the **Element Plan 1** tab for the **Dimensioning Points** parameter. When setting the **Min-Max Points**, there is no labeling for the corresponding symbol fixtures.

Precast slab

Calculations of shear force and bond

The results of the calculations of shear force and bond as per DIN EN 1992-1-1:2011-01 conducted using the  **Shear Reinforcement** tool had always been displayed in Internet Explorer, regardless of the default browser configuration.

This has now been changed. The configuration for the operating system's default browser will now generally be taken into account.

Precast Wall

Replaced functions for wall panels

Precast Wall = Wall·panel can no longer be selected in  **Design** in Allplan 2023. Please use the  **Design (iWall)** function for **Element types = Concrete wall** and **Sandwich wall** instead.

As a result, the special functions  **Angle for Assembly** in the **Fixtures** task area,  **Cast-in nuts for guardrail system** under  **Assembly Parts** and  **Secondary Reinforcement** for **Wall panel** type walls have been removed. Furthermore, the outlet of the function  **Modify Wall in View** has been changed slightly. The option to enter front mounted elements using  **Manually add front mounted element** is also omitted along with the **wall panel design mode** design type. The outlet after selecting  **Modify Wall in View** has therefore also been changed. Now, once you have selected this tool, the first wall is folded into the view window and you can also select one of the modification options ( **Modify wall element geometry**, etc.).

The **wall·panel** with the entries **Panel type catalog** and **Reinforcement type catalog** has been removed under  **Catalogs, Wall**. Likewise, the **Wall·panel** tab featuring special settings for the production of wall·panels under  **Catalogs, Site facilities, NC-Gen Driver catalog** is also omitted. With regard to the  **configurations** for the **wall programs**, the entries for the **wall·panel** (**Design** and **Reinforcement**) are omitted.

Should you have any further questions on this, please contact your personal sales engineer or our technical support team in Puch.

Consideration of wall-height architectural recesses in iWall

In the case of wall-height architectural recesses (generated, for example, using  **Door** or  **Joint**) which separate a wall completely, this process had to date been designed beyond the corresponding recess by using  **Spacing characteristic**.

This has now been improved. A new wall element is started after each wall-height recess separating the wall.

Ignore Precast element element state for all wall types in iWall

The **Element state = Ignore precast element**, which had to date only been available for walls generated using the  **Design (iWall)** function for **Composite sheeting** type walls, is now available for all wall types. Once this element state has been set, the corresponding precast elements (such as very short wall elements) are now ignored for all annotations.

To set, activate the desired element and manually set the element state on the **Attribute** tab in the **Element States and Actions** group.

- The corresponding element is consequently excluded from Rear-range Marks. It no longer occupies a position in the regular number range of the mark numbers.
- For the automatic labeling, a "----" as construction lines is displayed in place of the mark number. The text underneath the mark number or variable text image is also displayed as construction lines.
- The element is no longer displayed in the Element Plan. However, when the panel is being edited, the element may continue to be displayed in order to edit it using  **Model Precast Elements**.
- The corresponding precast element is excluded from all annotations, such as the generation of Production Data or lists, the TIM Export or the Element-plan-in-batch-run.
- This state can also be manually removed again too.

Sleeve shift on collision with lattice girders for iWall

When generating or modifying a wall generated using the  **Design (iWall)** tool, an automatic offset of the Cast-in nuts can be selected if they collide with the Lattice Girders. As a result, Allplan 2023 no longer features complex checks and reworking that had previously been necessary, for example, to prevent the cast-in nuts in the second leaf from being flattened by the Lattice Girders when the leaf produced first of all is turned in.

To define whether or not a check and therefore a subsequent offset of the cast-in nuts should occur, the **Cast-in Nut Offset** was expanded under  **Basic Reinforcement**,... on the **Cast-in Nuts** tab in the **Other Cast-in Nut Parameters** subdialog.

The **Consider Collisions with Lattice Girders = Yes/No** parameter (check box) had been added in the **Cast-in Nuts Offset** group. This check box is only visible for a **span direction** (setting on the  **Basic reinforcement**,... tab in the **Basic reinforcement general** group) of 0 ° (= Lattice Girders Perpendicular in Element). No setting can be made for any deviating values for the span direction. The offset of the Cast-in Nuts may need to take place manually, where required.

If the check box remains disabled, the algorithm will remain complete unchanged. If the checkbox is active, an **Offset to lattice girders** parameter appears. Here enter the required offset of the mounting part for the sleeves to set the lattice girder axis. The offset is always based on the enveloping min-max box of the fixture.

Note: In the following investigation, only those Lattice Girders that can be generated using the automatic features in iWall are investigated. These are the normal **Lattice Girders** and the **Intermediary carriers** for the **Concrete wall**, **Double wall** and **Thermal wall** Design Modes. In the case of the **Thermal wall**, the **auxiliary lattice girders** are also considered. Please note the aforementioned **Lattice Girders**, **Intermediate carriers** or **auxiliary lattice girders** in the event of a manual offset, copy, etc., or the manually generated secondary lattice girders, which have been integrated into the  **Element Plan** or the  **panel editing** using the  **Secondary Girders** or  **Shear Girders** tools in  **Secondary Reinforcement**, cannot be controlled.

If the **Consider Collision with Lattice Girders** option is active and Lattice Girders have been added in the corresponding element in the first place, a Collision Check and, where required, an offset of the Cast-in Nuts will be undertaken. The Cast-in Nuts will initially be positioned according to the existing rules. Checks are then made to see if the C-i nut is located within the sphere of influence of a Lattice Girder.

The sphere of influence of the Lattice Girder is defined by the value stipulated under **Offset to Lattice Girder Axis**. This is applied to the left and right next to the Lattice Girder axis.

If the C-i nut is not located within this sphere of influence, a valid position is therefore found for the C-i nut. If, however, a C-i nut is located within the sphere of influence specifically on or to the left or right next to the Lattice Girder axis, this must be offset further.

Note: It is only the Lattice Girder axis and the position of the fixture that will generally be investigated for the Cast-in Nuts. There is no real Collision Check undertaken between the Lattice Girders and the fixtures. In the case of flat Cast-in Nuts that do not collide with the Lattice Girder at all, the **Consider Collision with Lattice Girders** option must therefore be switched off where required.

The direction of the offset is in accordance with the configured variant for the **Offset of Cast-in Nuts** parameter.

- If the configured variant **Only horizontal** is selected, the C-i nut is therefore offset further in the direction (left or right) next to the Lattice Girder into a position where it is next to the Lattice Girder.

If the C-i nut is located precisely on the Lattice Girder axis, it will be offset outwards (to the left or right, i.e. away from the center of gravity).

Note: When selecting the **Center of gravity** option for **C-i nut arrangement**, the C-i nut will always be displaced to the right.

The C-i nut is therefore offset until the min-max box of the C-i nut is completely outside of the defined range.

- If **any** is the chosen variant, the C-i nut will initially be displaced horizontal, vertical or in any direction (depending on the shortest distance). If it is positioned within the sphere of influence of a Lattice Girder, it will be offset further until it is outside of the sphere

of influence.

However it will only be offset horizontally, analogous to the **Only horizontal** variant. As a result, the same rules regarding the direction of the offset and the min-max box of the C-i nut as those applicable to the **Only horizontal** variant will apply.

Finally, checks are made once again to see if the **Minimum spacing from openings** and **Minimum spacing from element edges** offsets in the **Edge offsets** group of the **Other Cast-In Nut Parameters** subdialog, which is found on the **Cast-In Nuts** tab under  **Basic reinforcement**,... have been observed.

If this condition is met, a valid position is therefore found for the C-i nut. If, on the other hand, the C-i nut slips too close to an opening, an element edge or one of the following, automatically generated Lattice Girders owing to the additional offset, it will be left in the original position before the additional offset, and the message **Minimum spacing of C-i nut from the Lattice Girder cannot be observed owing to a clash in spacing'** will appear. Please change the parameters output.

Improvements in input dialogs for iWall

On the **Division Parameters** under  **Spacing characteristics**, you can only select the entry for **Round Lengths** if either the **Consider Cranes** option under **Length Spacing Variants** or the **Limit Weight** option under **Length Spacing Parameters** have been selected beforehand. The entry for **Round Lengths** and the **Rounding value** were therefore positioned directly underneath the entries for **Limit Weight** in order to provide a better overview.

Under  **Basic reinforcement,...** on the **Lifting bolts** tab, the **Check lateral lifting bolts** option in the **Definition of lifting bolts** can now only be configured if the **Generate** option has been activated in the **Lateral lifting bolts** group.

Boolean functions for user-defined architectural elements in views/sections

Boolean functions can now also be applied in views and sections to precast elements, which have been generated from a User-Defined Archit. Element using the  **Precast Elements** or  **Design (iWall)** tools. This could, to date, only be applied in the Plan View or isometry.

Precast Elements

Removed functions in Precast Element task

The  **Axis Grid for Precast Elements**,  **Explode Axis Grid for Precast Elements** and  **Multi-Pitch Roof Plane** tools in the **Precast Element** task are no longer available in Allplan 2023. You must use the corresponding  **Axis Grid**,  **Explode Axis Grid** basic tools as well as the tool to model roof frames ( **Roof Frame** and  **Custom Planes**) instead.

Span direction as 3D vector

In the case of structural precast elements and precast elements, the span direction was previously defined by the 2D angle on the globe model. A span direction of 0° therefore always points east in relation to this globe. Special rules were laid down for the North and South Poles. This meant that a 2D entry angle (such as  **Entry via two points**) could have a completely different span direction than intended. Direct entry of the span direction as a 3D vector had not been possible to date. An auxiliary view had therefore always been required for this purpose.

This has now been changed. In the new version of Allplan, 3D vectors are defined that can be defined by two 3D points when entering the viewing and span direction in the **Orientation** group via two points. The 3D vector is then displayed for the viewing and span direction in place of the existing 2D vector. Furthermore, the two parameters **Viewing direction** and **Span direction** now have the same four, standard input options  **Entry via two points**,  **Entry via three points**,  **Entry via midpoint** and  **Switch direction**.

In the case of structural precast elements, the **Production dimensions** and **Loading dimensions** groups are also available alongside the **Orientation** group. Both of these groups have been adjusted accordingly.

Furthermore, when setting **absolute** for the **production site** or **loading site**, you can also choose from the four, standard input options

* Entry via two points, ** Entry via three points,  Entry via midpoint and  Switch direction, as entry here is now also via 3D points. The existing entry has been retained for the **Relative to span direction** setting. The existing entry via 2D points with the  Entry angle in Plan View, * Entry via two points and  Switch direction variants can all still be selected. In order to make reference to the entry via 2D points, the two **Production direction** und **Loading direction** parameters have been renamed here, becoming **Production direction 2D** and **Loading direction 2D**.

The entry sequence in the **Orientation** group has now been amended to **Reference point**, **Viewing direction** and **Span direction**.

Note: In the case of Precast Wall (iWall), the entry sequence has also been changed to **Reference point**, **Viewing direction**. As before, it is not necessary to enter the **Span direction** in the **Orientation** group for PythonParts and 3D objects.

Boolean functions for user-defined architectural elements in views/sections

Boolean functions can now also be applied in views and sections to precast elements, which have been generated from a User-Defined Archit. Element using the  **Precast Elements** or  **Design (iWall)** tools. This could, to date, only be applied in the Plan View or isometry.

Formwork

Shuttering boards in the Objects' palette

In the **Objects** palette, all shuttering boards had previously been listed in the **Shuttering board** group.

To improve the distinction between targeted visible and invisible switches and to determine the transparency or color of automatically generated and manually added shuttering boards, the shuttering boards are now listed separately in the **Automatic Shuttering Board** and **Manual Shuttering Board** groups.

Production Planning

Reinforcement groups in Production Planning

Previously, only **Reinforcement groups** for which the **MWS** entry option had been set for the selection in  **Production Data, NC Generator** could be selected and sent to Production. Previously, you could select the options **Standard** and **Expanded** for the **Bend definition** for transfer to production under  **Catalogs, Site facilities, NC generator driver catalog** on the **Filter** tab when selecting **Bar type = Reinforcement groups**.

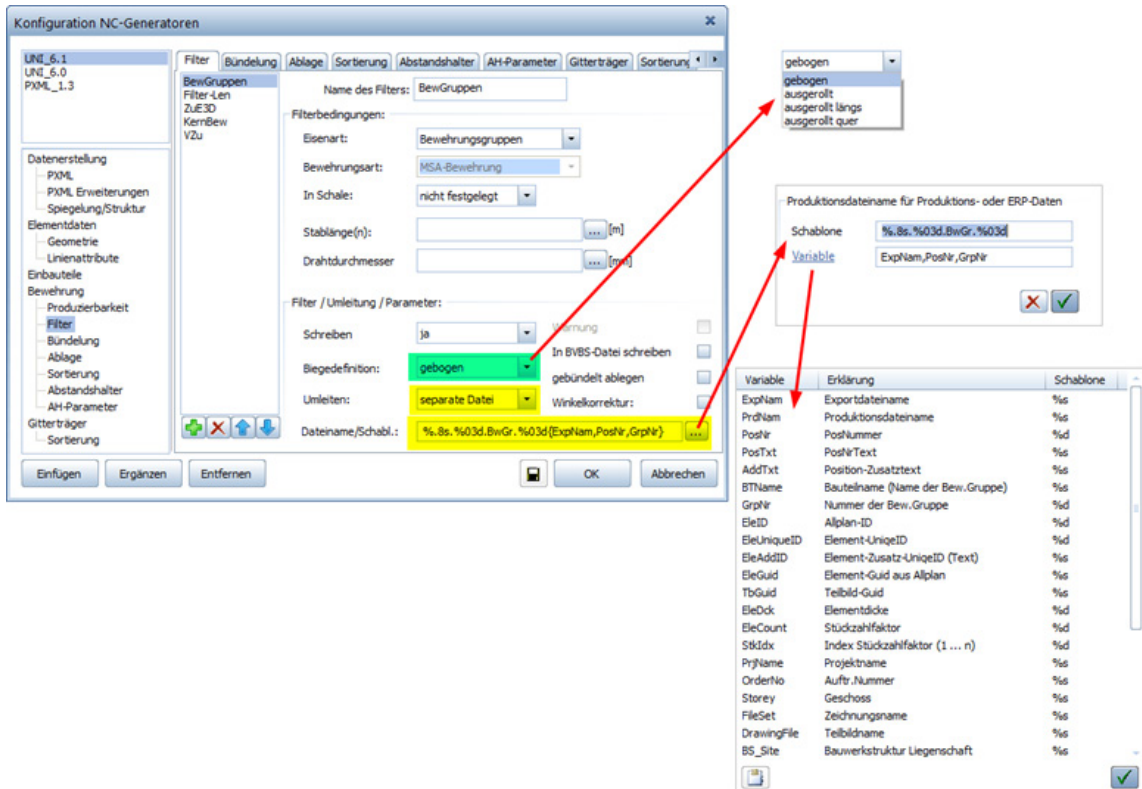
If **unfolded** was selected, the reinforcement group was then unfolded in the longitudinal direction and any crossbars attached to the segments were unfolded as well. However, bent crossbars were never unfolded.

This behavior has now been completely revised and enhanced.

- The tool can now be applied both to **Reinforcement groups** for which either the **Precast elements** entry option or the **MWS** entry option is set for the selection in  **Production Data, NC Generator**.
- Furthermore, 4 variants for the **Bend definition** are now available for selection, **bend**, **rolled out**, **rolled out LongIt** and **rolled out Cross**.
- Furthermore, **Reinforcement groups** in a precast element can now optionally be redirected to a separate file. To do this, select the new **Separate file** option under  **Catalogs, Site facilities, NC Generator Driver Catalog** on the **Filter** tab under **Redirect**. In this case, a **File name/Templ.** dialog (File name/Template) dialog will be displayed. By activating the sub-dialog, you can define a template for the file name of the reinforcement cage file, analogous to the production file name on the **Data creation** tab.

Clicking the  button behind the input field opens a sub-dialog where you can define the **Template** and the **Variable** for the creation of the file name. For this purpose, the new variable **GrpNr** = **Number of the reinforcement group** has been added to the

other variables. You can select the **component name (name of the reinforcement group)** via the variable **BTName**.



Note: For Unitechnik, you can rotate the meshes to their base position if necessary. Base layer refers to the plane in which the defined base section is located. However, this only applies to groups that are in a precast element, since freely transferred groups are rotated to their base position anyway. To activate the option, under **Catalogs, Site facilities, NC-Generator Driver Catalog** on the **Reinforcement** tab, select the **Rotate mesh in base layer** option, for the two variants **Stapl.reinf. + leveling + cutting** or **Stapled reinforcement**. The option is always active and grayed out for **Leveling + cutting system**.

Write Order Attribute in PXML Delegate file to NC data

If up to now the option **NC files contain info from delegate file** was selected for the parameter **Merge PXML delegate files** on the tab **PXML extensions** for an entry for generating a PXML file in  **Catalogs, Process planning, NC Generator Driver Catalog**, then up to now only the **element info** entries from the delegate file were transferred from each element to the element of the NC file (Unitechnik). The **Order** attributes generated by the NC generator itself remained untouched.

The **Order** attributes are now also transferred from the delegate file to the file for the NC generator. The following rules apply to this:

- If an attribute of the delegate file does not yet exist in the NC file (Unitechnik), it is appended.
- If an attribute of the delegate file already exists in the NC file (Unitechnik), the content is overwritten with the content of the delegate file.
- Attributes that were created in the NC file (Unitechnik) but not in the delegate file remain untouched in the NC file.

Transfer of open polygons in PXML file

According to the PXML interface description, a **Shape** object with several points (**Vertices**) always forms a closed polygon (without the last point being positioned on the first point, as is the case in Unitechnik). An open polyline is therefore mapped in such a way so that the complete path is retraced.

Trails had previously transferred via separate shape objects for each section. This does not go against the interface description, but this method makes it difficult (if not impossible) to clearly distinguish between connected trails (for linear fixtures) and other fixtures (such as polygons or symbol fixtures).

From Version 2023 upwards, open route sections are therefore passed via the interface. This may have an impact on third-party systems, insofar as these do not take account of the interface definition (e.g. for a length calculation).

Name of fixtures in List Generator

In List Generator, the text for naming fixtures (such as Isokorb® XT type K-U-F-M4-V1-REI120-CV50-LR200-X120-H250-7.1) is divided into several lines, unless it fits into the available column width.

Note: In the previous lists, any text longer than the column width stipulated in the list was no longer printed.

The program therefore searches for a "favourable" place where the name can be wrapped around. Blanks are preferred for the text wrap. If no blank exists, the search looks for specific special characters, e.g. a slash, a hyphen, a comma, etc.

The text is dynamically wrapped in the list. The result in the lists may look different depending on the column width (see, for example, the fixture list, the single panel configuration and the measurement sheet).

Export of NC data to the TIM

AllplanAs the most recent version, 2023 includes the export of NC data to TIM. In future, the NC Data will be generated directly in TIM.

When creating data for TIM using the  **Export TIM Data** tool, a note will appear stating that the tool will no longer be available in future versions of Allplan. This warning can be suppressed until the end of the program.

Please ensure therefore you configure the Production Data export to TIM using the **Regenerate Data – from Database** option. The data are then generated on TIM itself.

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